DIRE DAWA INSTITUTE OF TECHNOLOGY

Department of Computer Science — Course Outline

# Introduction to Distributed Systems

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| **Course Code** | COSC4038 | **Academic Year** | 2018 E.C. |
| **Program** | B.Sc. in Computer Science (Regular) | **Credit Hours** | 3 Credit Hours (5 ECTS) |
| **Year / Semester** | Fourth Year / Second Semester | **Contact Hours** | Lecture: 2h \| Lab: 3h \| Tutorial: 2h / wk |
| **Course Type** | Compulsory | **Prerequisites** | Operating Systems (COSC3023) Data Comm. & Networks (COSC2032) |
| **Instructor** | Mr. Abdurazak Kedir (M.Sc.) | | |

## 1. COURSE DESCRIPTION

A distributed system is a collection of independent computers connected through a network that cooperate to perform tasks or provide services to users as a unified system. Examples include the World Wide Web, networked file systems, DNS, and large-scale multiprocessor systems.

This course introduces the fundamental principles, concepts, techniques, algorithms, and practices used in the design and implementation of distributed systems. It takes a systems-oriented approach, emphasizing infrastructure software, communication, processes, naming, synchronization, consistency, replication, and fault tolerance.

The course combines theoretical study with practical implementation. Students will gain hands-on experience developing and experimenting with distributed-system concepts using programming languages and middleware technologies.

## 2. COURSE RATIONALE

Modern computing increasingly depends on systems in which multiple computers cooperate across networks. Web applications, cloud services, distributed databases, network services, enterprise applications, and many other computing environments rely on distributed-system principles.

Understanding distributed systems is therefore important for Computer Science students because distributed applications introduce problems that do not normally occur in a single-computer environment. These include communication delays, partial failures, synchronization, concurrency, naming, consistency, replication, and recovery. This course provides students with the theoretical foundation and practical skills required to design and implement basic distributed applications.

## 3. COURSE OBJECTIVES

### General Objective

To provide students with a comprehensive understanding of the fundamental principles and practical techniques of distributed systems.

### Specific Objectives

- Understand fundamental concepts, characteristics, and goals of distributed systems.
- Explain different distributed-system architectures and the role of processes, threads, clients, and servers.
- Examine communication mechanisms including RPC and remote object invocation.
- Explain naming mechanisms (flat, structured, attribute-based) and clock synchronization.
- Understand consistency, replication, fault tolerance, process resilience, and recovery mechanisms.
- Develop simple distributed applications using Java and appropriate middleware technologies.

## 4. LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- LO1 **Fundamentals:** Explain definitions, characteristics, goals, applications, and types of distributed systems.
- LO2 **Architectures:** Describe and analyze architectural styles and system architectures.
- LO3 **Processes:** Explain roles of processes, threads, clients, servers, and code migration.
- LO4 **Communication:** Implement major communication mechanisms including sockets, RPC, and RMI.
- LO5 **Naming:** Explain flat, structured, and attribute-based naming mechanisms.
- LO6 **Synchronization:** Apply physical and logical clock synchronization techniques.
- LO7 **Consistency & Replication:** Explain consistency requirements and scaling replication strategies.
- LO8 **Fault Tolerance:** Explain failure models, process resilience, distributed commit, and recovery.
- LO9 **Practical Development:** Develop basic distributed applications using Java and distributed middleware.
- LO10 **System Design:** Analyze distributed challenges and design scalable, transparent solutions.

## 5. COURSE CONTENT

### Chapter 1: Introduction to Distributed Systems (Allocated Time: 2 Hours)

**Topics:** 1.1 Introduction | 1.2 Definition | 1.3 Goals (Resource sharing, transparency, openness, scalability, reliability) | 1.4 Types of Distributed Systems.

**Expected Outcome:** Define distributed systems, explain goals, and distinguish common types.

### Chapter 2: Distributed System Architectures (Allocated Time: 4 Hours)

**Topics:** 2.1 Introduction | 2.2 Architectural Styles (Client-server, P2P, multi-tier, layered, middleware) | 2.3 System Architectures (Centralized vs. distributed, scalability design).

**Expected Outcome:** Explain architectural styles and analyze suitability for specific applications.

### Chapter 3: Processes in Distributed Systems (Allocated Time: 4 Hours)

**Topics:** 3.1 Introduction to Threads | 3.2 Threads in Distributed Systems | 3.3 Clients | 3.4 Servers (Stateless/stateful) | 3.5 Code Migration (Mobile code, resource/process migration).

**Expected Outcome:** Explain threads, clients, servers, and code migration in distributed environments.

### Chapter 4: Communication in Distributed Systems (Allocated Time: 6 Hours)

**Topics:** 4.1 Layer Protocols | 4.2 Types of Communication (Synchronous, asynchronous, socket communication) | 4.3 Remote Procedure Call (RPC stubs, parameter passing) | 4.4 Remote Object Invocation.

**Expected Outcome:** Explain communication mechanisms and implement basic inter-process communication.

### Chapter 5: Naming in Distributed Systems (Allocated Time: 4 Hours)

**Topics:** 5.1 Naming Entities (Names and identifiers) | 5.2 Flat Naming (Simple identifiers) | 5.3 Structured Naming (Hierarchical, naming services) | 5.4 Attribute-Based Naming.

**Expected Outcome:** Explain naming necessity and distinguish flat, structured, and attribute-based naming.

### Chapter 6: Synchronization in Distributed Systems (Allocated Time: 4 Hours)

**Topics:** 6.1 Overview | 6.2 Clock Synchronization | 6.3 Physical Clocks (Clock drift, synchronization algorithms) | 6.4 Logical Clocks (Lamport logical clocks, vector clocks, happened-before relationship).

**Expected Outcome:** Explain difficulties of time maintenance and apply physical/logical clock concepts.

### Chapter 7: Consistency and Replication (Allocated Time: 4 Hours)

**Topics:** 7.1 Introduction to Consistency | 7.2 Reasons for Replication (Performance, availability, reliability) | 7.3 Replication as a Scaling Technique (Replica management, trade-offs).

**Expected Outcome:** Explain replication use, consistency requirements, and scaling techniques.

### Chapter 8: Fault Tolerance (Allocated Time: 4 Hours)

**Topics:** 8.1 Basic Concepts | 8.2 Failure Models (Crash, communication failures) | 8.3 Process Resilience (Redundancy) | 8.4 Distributed Commit (Commit protocols) | 8.5 Recovery (Checkpointing, rollback).

**Expected Outcome:** Identify failures and explain resilience, commit, and recovery techniques.

## 6. LABORATORY COMPONENT

Using Java programming and distributed middleware environments:

| Lab | Laboratory Activity | Lab | Laboratory Activity |
|---|---|---|---|
| **Lab 1** | Implementation of Socket Programming | **Lab 7** | Multithreaded Client/Server Processes |
| **Lab 2** | Client-Server Program Using RMI | **Lab 8** | Program Demonstrating Code Migration |
| **Lab 3** | Client-Server Program Using RPC | **Lab 9** | Distributed Application Using EJB |
| **Lab 4** | Clock Synchronization (Logical/Physical) | **Lab 10** | CORBA Program Demonstrating Brokering |
| **Lab 5** | Implementation of an Election Algorithm | **Lab 11** | Distributed Application Using .NET Framework |
| **Lab 6** | Mutual Exclusion Algorithms | - | - |

## 7. SUGGESTED WEEKLY TEACHING SCHEDULE

| Week | Lecture / Tutorial Topic | Laboratory / Activity |
|---|---|---|
| 1 | Chapter 1: Introduction and definition | Orientation & Environment Setup |
| 2 | Chapter 1: Goals and types | Lab 1: Socket Programming |
| 3 | Chapter 2: Architectural styles | Lab 1 completion |
| 4 | Chapter 2: System architectures | Lab 2: RMI |
| 5 | Chapter 3: Threads and distributed processes | Lab 2 completion |

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| 6 | Chapter 3: Clients, servers and code migration | Lab 3: RPC |
| 7 | Chapter 4: Layer protocols and communication | Lab 4: Clock Synchronization |
| 8 | Chapter 4: RPC and remote object invocation | **Mid-Semester Examination** |
| 9 | Chapter 5: Naming entities and flat naming | Lab 5: Election Algorithm |
| 10 | Chapter 5: Structured and attribute-based naming | Lab 6: Mutual Exclusion |
| 11 | Chapter 6: Synchronization and physical clocks | Lab 7: Multithreaded Client/Server |
| 12 | Chapter 6: Logical clocks | Lab 8: Code Migration |
| 13 | Chapter 7: Consistency and replication | Lab 9: EJB Distributed Application |
| 14 | Chapter 7: Replication and scalability | Lab 10: CORBA |
| 15 | Chapter 8: Failure models, resilience and commit | Lab 11: .NET Distributed Application |
| 16 | Chapter 8: Recovery and course review | Project Presentation & Review |

## 8. TEACHING, LEARNING METHODS & ASSESSMENT

**Methods:** Lectures, interactive discussions, tutorial sessions, lab demonstrations, individual and group programming exercises, problem-solving, case studies, and course projects.

### Assessment Distribution

| Assessment Component | Weight |
|---|---|
| Assignments / Quizzes | 10% |
| Mid-Semester Examination | 20% |
| Course Project | 20% |
| Final Examination | 50% |
| **Total** | **100%** |

## 9. COURSE PROJECT & REFERENCES

**Project Areas:** Client-server information systems, distributed file-sharing apps, chat applications, library systems, inventory applications, database prototypes, and monitoring systems.

### Primary Textbook

Tanenbaum, A. S. and van Steen, M. *Distributed Systems: Principles and Paradigms*, 2nd Edition, Prentice Hall, 2007.

## References


- Coulouris, G. F., Dollimore, J., Kindberg, T. *Distributed Systems: Concepts and Design*, 5th Edition, Addison-Wesley, 2005.
- Ghosh, S. *Distributed Systems: An Algorithmic Approach*, 2nd Edition.
- Kshemkalyani, A. D. and Singhal, M. *Distributed Computing: Principles, Algorithms, and Systems*. Cambridge University Press, 2011.
- Attiya, H. and Welch, J. *Distributed Computing: Fundamentals, Simulations, and Advanced Topics*.
- Tel, G. *Introduction to Distributed Algorithms*.